SIMATS ENGINEERING
ASSESSMENT TOOL I

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I J. Daniel Jesuraj (192565085) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

J. Daniel
Signature of the student:

Q. No	*Understanding of Concepts (25)*	*Experimental Design (30)*	*Use of Tools (20)*	*Analysis of Results (15)*	*Documentation (10)*	*Total Score (out of 100)*
1	5 /5	5 /6	4 /4	3 /3	2 /2	19
2	5 /5	5 /6	4 /4	3 /3	2 /2	19
3	5 /5	5 /6	4 /4	3 /3	2 /2	19
4	5 /5	5 /6	4 /4	3 /3	2 /2	19
5	5 /5	5 /6	4 /4	3 /3	2 /2	19
OVERALL RUBRICS VALUE						
	5 /5	5 /6	4 /4	3 /4	2 /2	
	25 /25	25 /30	20 /20	15 /15	10 /10	95 /100

APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
Deduce the most likely original message.
Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
Demonstrate how the encryption process transforms the message step-by-step.
Explain how the repeating keyword influences each character of the ciphertext.

Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
Analyze how investigators could detect the presence of hidden data within the image.
Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CO-1 ASSESSMENT TOOL-1.

Name : Daniel Jesuraji .T

Reg No : 192565085

Course code : CSA5117

Course Name : cryptograph and Network security.

1) Event Communication security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing a message "MEET AT NOON" encoded using a shift pattern. how the encrypted message might appear if a shift of 4 positions is applied.

Plain text: MEET AT NOON

Shift: 4

Encryption:

Plaintext	M	E	E	T	A	T	N	O	O	N
Shift+4	Q	I	I	X	E	X	R	S	S	R

Ciphertext:

QIIX EX RSSR

Decryption:

The receiver shifts every letter 4 positions backward.

Example:

⇒ Q → M

⇒ I → E

⇒ X → T

Recovered plain text : MEET AT MOON.

Conclusion:-

The ceaser cipher encrypts data by shifting letters. The receiver decrypts the message using the same shift value (4).

2) Intercepted suspicious Message : A cybersecurity intern intercepts the coded message : "GSRH RH Z HVXIVG" during network monitering Deduce the most likely original message.

Cipher text:

GSRH RH Z HVXIVG

This is encrypted using the Atbash cipher;

where:

⇒ A ↔ Z

⇒ B ↔ Y

⇒ C ↔ X

⇒ D ↔ W

DECRYPTION:-

G → T
S → H
R → I
H → S
RH → IS
Z → A

HVXIVG ⟶ SECRET

Original Message:-

THIS IS A SECRET

Justification:-

⇒ The word RH repeatedly appears and becomes IS.

⇒ The last word becomes SECRET, a common english word.

⇒ Atbash is a monoalphabetic substitution cipher using reversed alphabets.

3) Secure messaging system a company deploys a keyword-based encryption technique for internal communication. A employee send "HELLO" using keyword "KEY" to ensure confidentiality.

Plaintext: HELLO

keyword : KEY

Repeat keyword:

Plain text	H	E	L	L	O
keyword	K	E	Y	K	E

Encryption:

using A=0, B=1, C=2 . . . . Z=25

Plain text	H	E	L	L	O
value	7	4	11	11	14
key	10	4	24	10	4
sum mod 26	17	8	9	21	18
cipher	R	I	J	V	S

cipher text:

RIJVS

Explanation:

The keyword repeats (KEYKE) and each letter shifts the plain text by the corresponding keyword value, making the encryption stronger than the ceaser cipher.

4) Suspicious Digital Asset investigation During a digital forensic audit, investigators suspect that company logo file may contain hidden information within it. Analyze how investigator could detect the presence of hidden data within the image.

Detecting hidden data:

Investigators can detect hidden information by:

1, statical analysis:

=> Examine pixel values for unusual patterns.

2. Steganalysis tools:-

=> use tools such as steg Expose or Steg

Solve to identify hidden data:

Two technique for hidden Improve security:-

=> Encrypt the secret message before embedding:

=> Even if extracted, the data remains un-readable without the decryption key,

=> Use Random Pixel Embedding.

=> Embed data in randomly selected pixels using a secret key instead of sequential pixel, making detection much better.

<u>conclusion:</u>

combining encryption with steganography Provides stronger confidentiality than using either technique alone.

5) Legacy encryption in Corporate communication (Rail Fence cipher).

cipher text:

WECRLTEERDSOEEFEAOCAIVDEN

a) original Message:

=> using Rail Fence cipher (3 rails)

=> The receiver knows that 3 rails were used. the cipher text is divided into the three trails and reconstructed using zigzag Pattern.

Recovered Plaintext:

WE ARE DISCOVERED FLEE AT ONCE

b) why is this method vulnerable:

=> it only changes the order of letters and does not alter the letter themselves

=> The frequency of letters remains unchanged, making cryptanalysis easier.

=> it does not provide sufficient security for modern communication systems.

c) Practical enhancement:

Use a modern encryption algorithm such as AES (Advanced Encryption standard) before transmitting the message. the AES uses a secret key and provides much stronger security than classical transposition ciphers.

Conclusion:

The rail Fence cipher is suitable only for learning purpose. Modern encryption algorithm such as AES provides far better confidentiality and protection against attacks.